

HEMP HOUSE



Why is this project a best practice?

Challenge

The project had to provide high thermal comfort through cold winters and warm summers while sharply reducing operational energy. The main design challenge was to coordinate a high-performance envelope, efficient services, renewable energy and healthy materials without compromising the architectural concept, construction quality or everyday usability.

Solution

The response combines hemp-lime used in walls and as insulation for walls, roof and doors; a hygroscopic envelope intended to regulate indoor humidity and reduce mould and pest risk; and local natural materials, laminated timber structure and recycled or locally made timber components. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The core package - demand reduction first, airtight construction, efficient low-temperature systems, ventilation and on-site renewables - is transferable to other Romanian homes. Replication requires climate-specific energy modelling, careful thermal-bridge and airtightness detailing, qualified installation, commissioning and clear operating guidance for residents.

Key facts and figures

Location	Sibiel, Sibiu County, Romania
Building use / function	Single-family model house
Project type (new build, refurbishment, extension)	New build / material demonstrator
Plot area / Floor area	Approximately 100 m ²
Year / phase	Model-house completion scheduled for 2019 in the RoGBC toolkit

Climate zone	Humid continental / cool Transylvanian climate
Thematic focus / innovation	Hemp-lime construction; natural local materials; moisture regulation; low embodied carbon
Investor / building owner	Stonehemp / private project
Architect	Not publicly disclosed in the consulted sources
Other involved stakeholders	Stonehemp; local timber and joinery suppliers; Romania Green Building Council
Contact person	Stonehemp / RoGBC
Further information	https://www.rogbc.org/proiecte-certificate-rogbc?lang=en https://www.rogbc.org/_files/ugd/40ee3e_b5d91d7a3b71422cb98cd7a289b89c79.pdf

Hemp House is a new build / material demonstrator project in Sibiel, Sibiu County, Romania. It functions as single-family model house. Its main best-practice contribution is A Romanian demonstration of bio-based construction in which the envelope stores carbon and supports healthy indoor conditions. The documented strategy brings together hemp-lime used in walls and as insulation for walls, roof and doors, a hygroscopic envelope intended to regulate indoor humidity and reduce mould and pest risk, and local natural materials, laminated timber structure and recycled or locally made timber components. The toolkit states that lime can absorb CO₂ during carbonation and that hemp-lime production uses less water; no verified whole-building LCA was publicly available. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - Hemp-lime used in walls and as insulation for walls, roof and doors.
2. Integrated systems and operation - A hygroscopic envelope intended to regulate indoor humidity and reduce mould and pest risk.
3. Wider value - Local natural materials, laminated timber structure and recycled or locally made timber components.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor

spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.